

Alisander Qoshqosh

Software Engineer

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About Me

Seasoned and self-directed software engineer, experienced with open-source, startup's MVPs and high load enterprise solutions. Advocate of **functional programming** and **KISS** design principle. Passionate about **Rust** programming language. Possess mathematical thinking, love math and cryptography. Used to work in a fast-paced and deadline-driven environment. Highly rate democratic processes in teams. Love car travels and snowboarding.

EXPERIENCE

Senior Developer (open source)

OpenZeppelin, Mar 2024 – Present, www.openzeppelin.com, blockchain security and infrastructure.

- Shipped the first open-source smart contracts [library](#) for Arbitrum [Stylus VM](#), that includes [ERC20](#), [ERC721](#), [ERC1155](#), [proxies](#) and others. Assisted with [security audit](#) - 3 audits passed successfully.
- Built the first [test framework](#) for Stylus VM.
- Lead development of on-chain efficient crypto library, that includes [big integers](#), efficient Montgomery based [finite fields](#), ZK-friendly [Poseidon hash](#) and [Pedersen hash](#), [EDDSA](#) signature and others.
- Worked with community: [speaker](#) on EthCC`25 (Cannes), [speaker](#) on ArbiVerse`24 (Bangkok), author of efficient Poseidon hash [article](#).

Rust | EVM | Solidity | WASM | L2 | Arbitrum Stylus | GitHub actions

Software Engineer (cryptography)

Gridlock, Jan 2023 – Mar 2024, www.gridlock.network, decentralized crypto wallet.

- Developed cryptographic rust library for **MPC** transaction signing, generation and recovery of distributed private keys with threshold (based on Gennaro-Goldfeder 2020 [white paper](#)).
- Introduced 2fa secrets recovery support (based on **Shamir's secret sharing**).
- Kept a keen eye on security: fixed a **zero-day vulnerability** issue CVE-2023-33241 for MPC wallet.
- Built CI: fulfilling integration tests and mpc node's compatibility checks with **github actions**.

Rust | NATs | MPC | EcDSA | EdDSA | WASM | TypeScript | MongoDB | GitHub actions

Software Engineer II

Kraken, Payward inc., May 2022 – Dec 2022, www.kraken.com, crypto exchange.

- (NDA)

Rust | Tokio | k8s | Redis | Kafka | MariaDB | GitLab

Senior Software Engineer

VEON, Oct 2020 - Apr 2022, www.veon.com, network/telecommunication provider, ~160 million clients.

- Designed and implemented backend for 2021 **best-selling** product (advertised as “Your Decision”) for the “beeline” main [website](#).
- Lead major architectural changes: split of monolithic system and **rust** adoption.
- Conducted interviews and mentored junior developers.

C#/.Net | Rust | Actix/Tokio | k8s | Redis | Java | Kafka | Elasticsearch | Oracle | MongoDB | GitLab

Software Engineer

Mobile TeleSystems, Apr 2019 - Oct 2020, www.mts.ru, network/telecommunication provider, ~80 million clients.

- Developed role-based **security auth** service and its client library.
- Built a custom serialization for the WCF RPC framework, reducing SOAP message size by **40%** and cutting overall backend latency by over **2x** for heavy queries.

C#/.Net | Oracle | PostgreSQL | PL/SQL | k8s | Redis | gRPC | Azure Service Bus | Azure | Azure DevOps

Software Engineer

Ibicom, Moscow, Jun 2017 - Apr 2019, system integration.

- Implemented main features for **JavaFX** desktop software for indoor navigation in real-time.

Java/JavaFX | Kotlin | AWS | MySQL

EDUCATION

Moscow Institute of Physics and Technology (MIPT)

Applied Mathematics and Control, Moscow, www.eng.mipt.ru